Feiyu Jia

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RESEARCH INTERESTS

Efficient machine learning systems for large language models (LLMs): extreme low-bit quantization and compression, efficient inference and serving (speculative decoding, multi-node serving), and algorithm–hardware co-design.

EDUCATION

New York University, Tandon School of Engineering — New York, NY
M.S. in Computer Engineering · GPA: 4.0/4.0 — Sep 2024 – May 2026

- Coursework: Deep Learning, Advanced Machine Learning, Computing Systems Architecture, Foundations of Robotics.

The Hong Kong University of Science and Technology (HKUST) — Hong Kong SAR
B.Sc. in Data Science and Technology — Sep 2020 – Jul 2024

University of Florida, Herbert Wertheim College of Engineering — Gainesville, FL
Exchange Student — Jan 2023 – May 2023

PUBLICATIONS

Under Review

[U1] CoBIT: Unified Optimization of Codebooks, Assignments, and Activation Composition for Extreme Low-Bit LLM Quantization
Xiangyang Yin, **Feiyu Jia**, Henry Kam, Tianhua Xia, Qi Sun, Sai Qian Zhang
Under review at the International Conference on Learning Representations (ICLR), 2027. [OpenReview]

[U2] DSD-Sim: A Simulation Framework for Distributed Speculative Decoding in Multi-Node LLM Serving
Leshu Li, Fengze Yu, Yifei Feng, **Feiyu Jia**, Jian Li, Hao Wang, Bradley McDanel, Sai Qian Zhang
Under review at the AAAI Conference on Artificial Intelligence (AAAI), 2027.

Conference Proceedings — * Equal contribution

[C1] Video Face Swap with DeepFaceLab
Feiyu Jia* (corresponding author), Sihan Yang*
Proc. SPIE 12168, International Conference on Computer Graphics, Artificial Intelligence, and Data Processing (ICCAID 2021), 121681H, 2022. [DOI]

RESEARCH EXPERIENCE

Research Assistant, New York University — Apr 2026 – Present
Systems and Artificial Intelligence (SAI) Lab · Advisor: Prof. Sai Qian Zhang — New York, NY
Efficient LLM inference: extreme low-bit quantization and distributed speculative decoding

- **CoBIT [U1].** Co-developed a post-training quantization framework that jointly optimizes Hessian-aware weight clustering (codebooks + assignments) and per-token FP4 exponent–mantissa selection (PhaseLUT).
- Implemented and ran experiments against five baselines: on Qwen3 (4B–32B), +4.8 / +35.9 points average accuracy over the strongest baseline at A4W3 / A4W2, and the best average on three reasoning LLMs (A8W4); at A4W4 the fused CUDA kernel is 1.64–3.66× faster than FP16 cuBLAS, up to 1.53× end to end.
- **DSD-Sim [U2].** Co-developed a discrete-event simulator (SimPy + VIDUR) of multi-node speculative decoding: draft–target coordination, token acceptance, routing, batching, speculation windows, network topology.
- Calibrated against A40/A100/H100 GPUs (3.2%/4.4% decode/prefill latency error) and NS-3 (≈3% error) up to 4,096 nodes; showed JSQ routing's edge reverses near saturation and length-aware batching mainly cuts latency.

Research Assistant, AI Subteam, NYU Vertically Integrated Projects (VIP) — Jan 2025 – Jul 2025
Smart Microphysiological System for Precision Medicine · Advisor: Prof. Weiqiang Chen — New York, NY
Deep learning-guided immunotherapy screening for pancreatic adenocarcinoma

- Built an automated pipeline to detect, count and track cancer and immune cells in time-lapse microscopy image sequences, comparing a ConvLSTM tracker with an OpenCV contour-based detector.

- Developed and benchmarked deep survival models on public TCGA histopathology images: EfficientNet and Swin Transformer backbones against DeepSurv and Random Survival Forests on CNN-extracted features.
- Annotated a new microscopy dataset (cancer cells, immune cells, vasculature) with a custom Python labeling script, and tested classical trackers (Lucas–Kanade optical flow, KCF, CSRT) for speed and robustness.

Undergraduate Researcher (UROP), HCI Initiative, HKUST Jul 2022 – Dec 2022
Advisor: Prof. Xiaojuan Ma · Department of Computer Science and Engineering Hong Kong SAR
Assessing user experience to design effective visual representation and interaction in virtual reality

- Built Unity (C#) VR prototypes for Meta Quest 2 with an immersive analytics toolkit (IATK) for data visualization and the Meta Interaction SDK for hand tracking: distance grab, poke/scroll and hand-pose recognition.
- Conducted a user study with 32 participants on the accuracy, operability and scalability of hand-gesture interaction for immersive data exploration.
- Identified limits of current hand tracking (orientation-sensitive pose detection, no multi-finger poke) and documented setup guidelines and design implications in a technical report.

Independent Research Project: Video Face Swapping, HKUST 2021

- Trained a SAEHD autoencoder with S3FD scale-invariant face detection for frame-by-frame face replacement in video; led data collection, training, evaluation and writing, resulting in a co-first-author SPIE paper [C1].

SELECTED PROJECTS

EcoTraining: Energy-Efficient Gradient Compression Hardware for LLM Training Feb 2026
Cognichip Hackathon · Team "Train Neo Bit" (4 members) [GitHub]

- Co-designed in SystemVerilog a hierarchical on-chip gradient compressor: large gradients bypass directly, small ones accumulate in a set-associative SRAM cache until they cross a threshold, and a FIFO batches DRAM write bursts; cut DRAM writes by 52% (2.09× compression) in testbench simulation.

INDUSTRY EXPERIENCE

Product Operations Intern, Tencent Jul 2025 – Oct 2025
Tencent Technology (Shenzhen) Co., Ltd. Shenzhen, China

- Supported daily operations of CodeBuddy, Tencent's AI coding assistant, including data workflows and quality optimization; analyzed user data to identify issues and support product iteration.
- Coordinated with managers and engineering teams on the deployment and optimization of AI features.

SKILLS

Programming	Python, C++, CUDA, C#, Java, R, SQL, MATLAB
ML & Systems	PyTorch, TensorFlow/Keras, OpenCV, SimPy, Linux, Git
Hardware & XR	SystemVerilog/Verilog, Unity, Meta Quest / XR Interaction SDK

REFERENCES

Prof. Sai Qian Zhang sai.zhang@nyu.edu
Assistant Professor, ECE (Tandon) and Computer Science (Courant), New York University